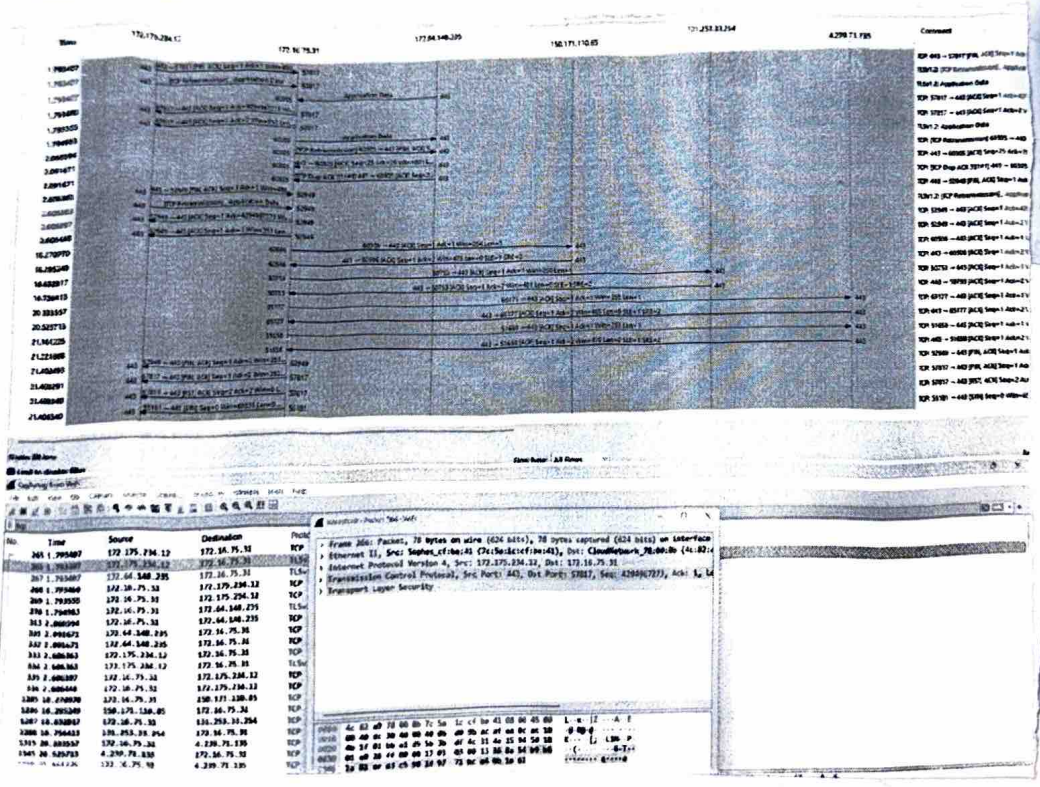
Wireshark · Display Filters

Filter Name	Filter Expression
Ethernet address 00:00:5e:00:53:00	eth.addr == 00:00:5e:00:53:00
Ethernet type 0x0806 (ARP)	eth.type == 0x0806
Ethernet broadcast	eth.addr == ff:ff:ff:ff:ff:ff
No ARP	not arp
IPv4 only	ip
IPv4 address 192.0.2.1	ip.addr == 192.0.2.1
IPv4 address isn't 192.0.2.1	ip.addr != 192.0.2.1
IPv6 only	ipv6
IPv6 address 2001:db8::1	ipv6.addr == 2001:db8::1
TCP only	tcp
UDP only	udp
Non-DNS port	!(udp.port == 53 || tcp.port == 53)
TCP or UDP port is 80 (HTTP)	tcp.port == 80 || udp.port == 80
HTTP	http
No ARP and no DNS	not arp and not dns
Non-HTTP and non-SMTP to/from 192.0.2.1	ip.addr == 192.0.2.1 and tcp.port not in {80, 25}



Exp no: 5 Experiment on packet capture tool Wireshark

Aim

To understand the feature of wireshark as a packet capture tool and understand encapsulation of information at various layers of a protocol stack

Packet sniffer

* Capture packets sent and received by the computer
* Display protocol information of captured packets
* Works passively
* Receives copies of networks packets for analysis

packet sniffer tools:

* TCP dump - command line packet
* Wireshark - Graphical packet analyzer

Uses of Network:

Capture network Traffic

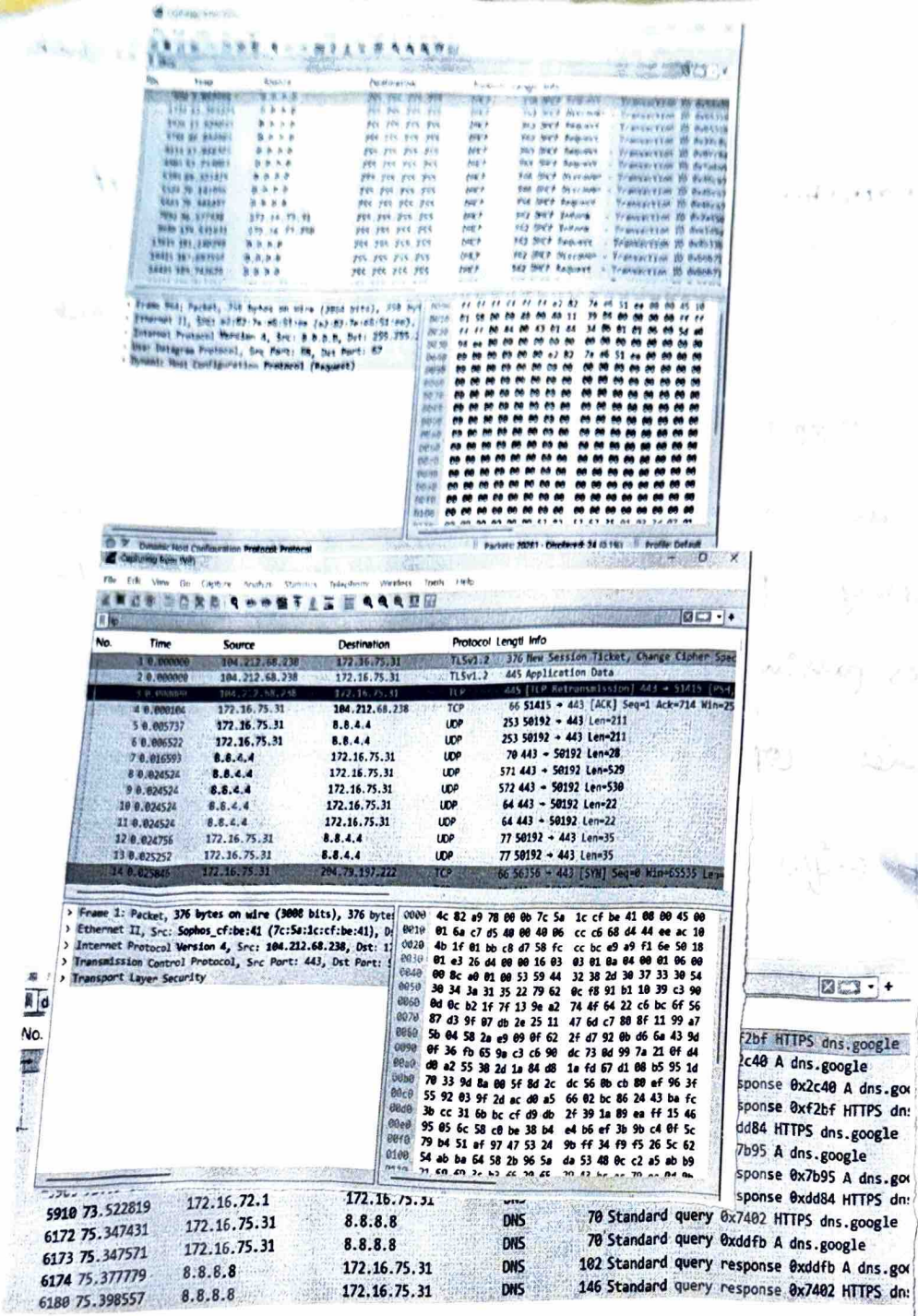
Analyzes protocol.

Flow graph:

Displays the communication flow between devices making packing

Capture 1000 packets:

* Open the capture file
* Apply the TCP || UDP filter
* Inspect and view the flow graph.
* Apply the DNS filter



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	104.212.68.238	172.16.75.31	TLSv1.2	376	New Session Ticket, Change Cipher Spec
2	0.000000	104.212.68.238	172.16.75.31	TLSv1.2	445	Application Data
3	0.000000	104.212.68.238	172.16.75.31	TCP	445	[TCP Retransmission] 443 → 51415 [PSH
4	0.000104	172.16.75.31	104.212.68.238	TCP	66	51415 → 443 [ACK] Seq=1 Ack=714 Win=25
5	0.005737	172.16.75.31	8.8.4.4	UDP	253	50192 → 443 Len=211
6	0.006522	172.16.75.31	8.8.4.4	UDP	253	50192 → 443 Len=211
7	0.016593	8.8.4.4	172.16.75.31	UDP	70	443 → 50192 Len=28
8	0.024524	8.8.4.4	172.16.75.31	UDP	571	443 → 50192 Len=529
9	0.024524	8.8.4.4	172.16.75.31	UDP	572	443 → 50192 Len=530
10	0.024524	8.8.4.4	172.16.75.31	UDP	64	443 → 50192 Len=22
11	0.024524	8.8.4.4	172.16.75.31	UDP	64	443 → 50192 Len=22
12	0.024756	172.16.75.31	8.8.4.4	UDP	77	50192 → 443 Len=35
13	0.025252	172.16.75.31	8.8.4.4	UDP	77	50192 → 443 Len=35
14	0.025845	172.16.75.31	204.79.197.222	TCP	66	56356 → 443 [SYN] Seq=0 Win=65535 Len

```
> Frame 1: Packet, 376 bytes on wire (3008 bits), 376 byte
> Ethernet II, Src: Sophos_cf:be:41 (7c:5a:1c:cf:be:41), D
> Internet Protocol Version 4, Src: 104.212.68.238, Dst: 1
> Transmission Control Protocol, Src Port: 443, Dst Port:
> Transport Layer Security
```

```
0000  4c 82 a9 78 00 0b 7c 5a  1c cf be 41 08 00 45 00
0010  01 6a c7 d5 40 00 40 06  cc c6 68 d4 44 ee ac 10
0020  4b 1f 01 bb c8 d7 58 fc  cc bc e9 a9 f1 6e 50 18
0030  01 e3 26 d4 00 00 16 03  03 01 8a 04 00 01 06 00
0040  00 8c a0 01 00 53 59 44  32 38 2d 30 37 33 30 54
0050  30 34 3a 31 35 22 79 62  0c f8 91 b1 10 39 c3 90
0060  0d 0c b2 1f 7f 13 9e a2  74 4f 64 22 c6 bc 6f 56
0070  87 d3 9f 07 db 2e 25 11  47 6d c7 80 8f 11 99 a7
0080  5b 04 58 2a e9 09 0f 62  2f d7 92 0b d6 6a 43 9d
0090  0f 36 fb 65 9a c3 c6 90  dc 73 8d 99 7a 21 0f d4
00a0  d0 a2 55 38 2d 1a 84 d8  1a fd 67 d1 08 b5 95 1d
00b0  70 33 9d 8a 00 5f 8d 2c  dc 56 0b cb 80 ef 96 3f
00c0  55 92 03 9f 2d ac d0 a5  66 02 bc 86 24 43 ba fc
00d0  3b cc 31 6b bc cf d9 db  2f 39 1a 89 ea ff 15 46
00e0  95 05 6c 58 c0 be 38 b4  e4 b6 ef 3b 9b c4 0f 5c
00f0  79 b4 51 af 97 47 53 24  9b ff 34 f9 f5 26 5c 62
0100  54 ab ba 64 58 2b 96 5a  da 53 48 0c c2 a5 ab b9
```

No.					
					f2bf HTTPS dns.google
					2c40 A dns.google
					sponse 0x2c40 A dns.goo
					sponse 0xf2bf HTTPS dns
					dd84 HTTPS dns.google
					7b95 A dns.google
					sponse 0x7b95 A dns.goo
					sponse 0xdd84 HTTPS dns
5910	73.522819	172.16.72.1	172.16.75.31		
6172	75.347431	172.16.75.31	8.8.8.8	DNS	70 Standard query 0x7402 HTTPS dns.google
6173	75.347571	172.16.75.31	8.8.8.8	DNS	70 Standard query 0xddfb A dns.google
6174	75.377779	8.8.8.8	172.16.75.31	DNS	102 Standard query response 0xddfb A dns.goo
6180	75.398557	8.8.8.8	172.16.75.31	DNS	146 Standard query response 0x7402 HTTPS dns

```
> Frame 5047: Packet, 70 bytes on wire (560 bits), 70 bytes
> Ethernet II, Src: CloudNetwork_78:00:0b (4c:82:a9:78:00:0
> Internet Protocol Version 4, Src: 172.16.75.31, Dst: 172.
> User Datagram Protocol, Src Port: 53918, Dst Port: 53
> Domain Name System (query)
```

```
0000  7c 5a 1c cf be 41 4c 82  a9 78 00 0b 08 00 45 00  |z
0010  00 38 65 85 00 00 80 11  e9 ee ac 10 4b 1f ac 10  ·8
0020  48 01 d2 9e 00 35 00 24  99 73 f2 bf 01 00 00 01  H
0030  00 00 00 00 00 00 03 64  6e 73 06 67 6f 6f 67 6c  ·
0040  65 00 00 41 00 01                                e
```

arp

No.	Time	Source	Destination	Protocol	Length	Info
1343	10.957901	CloudNetwork_78:90:...	Broadcast	ARP	42	Who has 172.16.75.125? Tell 172.16.75.6
1977	14.358615	CloudNetwork_78:90:...	Broadcast	ARP	42	Who has 172.16.77.18? Tell 172.16.75.61
2098	15.471882	CloudNetwork_78:90:...	Broadcast	ARP	42	Who has 172.16.77.18? Tell 172.16.75.61
2400	17.952584	CloudNetwork_78:90:...	Broadcast	ARP	42	Who has 172.16.78.5? Tell 172.16.75.61
2747	24.883561	CloudNetwork_78:90:...	Broadcast	ARP	42	Who has 172.16.75.204? Tell 172.16.75.6
4906	61.332363	CloudNetwork_77:ff:...	Broadcast	ARP	42	Who has 172.16.77.93? Tell 172.16.75.53

* Inspect and write flow graph and save the Capture.

Result:

Thus the feature of wireshark and implement of different filter was successfully completed